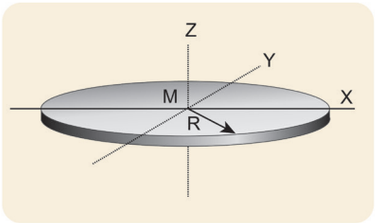
(iii) The moment of inertia (I) about an axis passing through the centre and lying on the plane of the disc is,



$$I_Z = I_X + I_Y$$

where, $I_X = I_Y = I$ and $I_Z = \frac{1}{2}MR^2$

$$I_Z = 2I;\ I = \frac{1}{2}I_Z$$

$$I = \frac{1}{2}\times\frac{1}{2}MR^2 = \frac{1}{4}MR^2$$

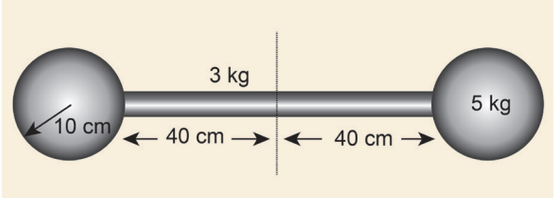
$$I = \frac{1}{4}\times 3\times(0.5)^2 = 0.25\times 3\times 0.5\times 0.5$$

$$I = 0.1875\,\text{kg m}^2$$

- About which of the above axis it is easier to rotate the disc?
- It is easier to rotate the disc about an axis about which the moment of inertia is the least. Hence, it is case (iii).

EXAMPLE 5.17

Find the moment of inertia about the geometric centre of the given structure made up of one thin rod connecting two similar solid spheres as shown in Figure.



Solution

The structure is made up of three objects; one thin rod and two solid spheres.

The mass of the rod, M = 3 kg and the total length of the rod, ℓ = 80 cm = 0.8 m

The moment of inertia of the rod about its centre of mass is, $I_{rod} = \frac{1}{12}M\ell^2$

$$I_{rod} = \frac{1}{12}\times 3\times(0.8)^2 = \frac{1}{4}\times 0.64$$

$$I_{rod} = 0.16\,\text{kg m}^2$$

The mass of the sphere, M = 5 kg and the radius of the sphere, R = 10 cm = 0.1 m

The moment of inertia of the sphere about its centre of mass is, $I_C = \frac{2}{5}MR^2$

The moment of inertia of the sphere about geometric centre of the structure is, $I_{sph} = I_C + Md^2$

Where, d = 40 cm + 10 cm = 50 cm = 0.5 m

$$I_{sph} = \frac{2}{5}MR^2 + Md^2$$

$$I_{sph} = \frac{2}{5}\times 5\times(0.1)^2 + 5\times(0.5)^2$$

$$I_{sph} = (2\times 0.01) + (5\times 0.25) = 0.02 + 1.25$$

$$I_{sph} = 1.27\,\text{kg m}^2$$

As there are one rod and two similar solid spheres we can write the total moment of inertia (I) of the given geometric structure as, $I = I_{rod} + (2\times I_{sph})$

$$I = (0.16) + (2\times 1.27) = 0.16 + 2.54$$

$$I = 2.7\,\text{kg m}^2$$

5.4.6 Moment of Inertia of Different Rigid Bodies

The moment of inertia of different objects about different axes is given in the Table 5.3.

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